Arista Access Switch Node IND-ACC-65 Indiranagar Exchange Minor QoS/Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Root Cause Isolation and Resolution SOP

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5454

Description

The occurrence of QoS/congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on the Arista Access switch node IND-ACC-65 at the Indiranagar Exchange site can be attributed to the intricate interplay between the forwarding and queuing mechanisms, as well as the traffic policing and shaping functionalities of the switch. In the context of this specific role and vendor equipment, the issue may arise due to an imbalance in the configured QoS policies, an over-subscription of available bandwidth by connected devices, or the activation of certain hardware or software features that inadvertently exacerbate congestion. Furthermore, the switch's reliance on shared buffers and the dynamic nature of traffic patterns can also contribute to the manifestation of these events. Consequently, it is essential to thoroughly examine the QoS configuration, verify the operational status of the interfaces, and assess the impact of any configured policing or shaping policies to accurately diagnose and resolve the root cause of this issue.

Pre-requisite Checks

- prerequisite: Verify the availability and accessibility of the Indiranagar Exchange site, including the remote hands team, to facilitate any necessary physical interventions.

- prerequisite: Ensure that both the primary and secondary links to the affected switch node are functioning correctly and that any redundant link status is verified.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to an imbalance in bandwidth allocation.	High
Over-subscription of available bandwidth by connected devices.	Medium
Inadvertent activation of hardware or software features that exacerbate congestion	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policer all, Expected Output: A comprehensive list of configured QoS policing policies, including their priority levels, rates, and burst sizes., Description: This command provides an overview of the configured QoS policing policies, allowing for the identification of any potential misconfigurations or anomalies that may be contributing to the QoS/congestion event.
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```
Command: show interface all counters, Expected Output: A detailed list of interface counters, including the number of packets and bytes transmitted and received, as well as errors and discards., Description: This command provides a detailed view of interface activity, enabling the identification of any congested interfaces or potential issues with packet transmission or reception.

Command: show queue all detail, Expected Output: A detailed list of queue information, including the current queue depth, maximum queue depth, and average queue depth., Description: This command provides a detailed view of queue activity, enabling the identification of any congested queues or potential issues with queue management.
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Resolution Steps

Step 1: Step: Verify and adjust QoS policies as necessary., Command: config qos policer rate burst , Description: Adjust the QoS policing policies to ensure that they are correctly configured and aligned with the expected traffic patterns and requirements.

Step 2: Step: Verify and adjust interface settings as necessary., Command: config interface speed , Description: Adjust the interface settings to ensure that they are correctly configured and aligned with the expected traffic patterns and requirements.

Step 3: Step: Verify and adjust queue settings as necessary., Command: config queue max-depth , Description: Adjust the queue settings to ensure that they are correctly configured and aligned with the expected traffic patterns and requirements.

Post-Check Verification

```
Command: show qos policer all, Expected Output: A comprehensive list of configured QoS policing policies, including their priority levels, rates, and burst sizes., Description: Verify that the QoS policing policies have been correctly configured and are aligned with the expected traffic patterns and requirements.

Command: show interface all counters, Expected Output: A detailed list of interface counters, including the number of packets and bytes transmitted and received, as well as errors and discards., Description: Verify that the interface activity is within expected ranges and that there are no congested interfaces or potential issues with packet transmission or reception.

Command: show queue all detail, Expected Output: A detailed list of queue information, including the current queue depth, maximum queue depth, and average queue depth., Description: Verify that the queue activity is within expected ranges and that there are no congested queues or potential issues with queue management.
```

Roll-back Procedures

- Step: Restore original QoS policies., Command: config qos policer rate burst , Description: Restore the original QoS policing policies to their previous configuration.

- Step: Restore original interface settings., Command: config interface speed , Description: Restore the original interface settings to their previous configuration.

- Step: Restore original queue settings., Command: config queue max-depth , Description: Restore the original queue settings to their previous configuration.